SMART WALKING STICK FOR VISUALLY IMPAIRED NAVIGATION WITH AI INTEGRATION

25-26J-270

Project Proposal Report

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

Independent navigation for visually impaired individuals depends on accurate awareness of both environmental landmarks and immediate physical hazards. Traditional white canes and single sensor electronic travel aids often deliver late or ambiguous warnings, trigger frequent false alarms, and cannot reliably distinguish between a safely passable gap and a blocked path or detect overhead and drop off hazards early enough. This component of the Smart Walking Stick project introduces a multi modal obstacle and gap detection system that complements the landmark recognition module by concentrating on near field physical safety. The design combines dual ultrasonic sensors for precise short range distance profiling with an ESP32 CAM module that supplies structural visual cues. Engineered temporal and spatial features derived from distance symmetry, variance, slope trends, edge density, scene occupancy, brightness, and vision confidence are processed by a compact quantized TensorFlow Lite for Microcontrollers model to classify states of safe, caution, or unsafe (including drop off and overhead conditions in extended states). A fusion and decision layer applies hysteresis, override rules for imminent collision or drop off conditions, uncertainty gating, and calibrated confidence thresholds to suppress flickers and reduce spurious alerts. Distinct vibrotactile patterns convey hazard type and severity while limiting cognitive load and avoiding alert fatigue. Power aware duty cycling dynamically lowers camera frame rate during stable safe periods and increases sensing density during ambiguous transitions, supporting sustained operation without sacrificing responsiveness. Anticipated outcomes include achieving an F1 score of at least 0.88 for unsafe versus safe discrimination, reducing false alarms by at least thirty percent compared with a simple ultrasonic threshold baseline, maintaining end to end response latency not exceeding three hundred fifty milliseconds at the ninety fifth percentile, and delivering a minimum of five hours continuous operation under a representative mixed usage profile. The system preserves privacy by avoiding raw image storage and remains cost effective with readily available low-cost hardware. Expected contributions encompass a reproducible multi modal tiny machine learning pipeline, a confidence aware haptic feedback scheme, a structured evaluation dataset specification, and a comparative assessment methodology for assistive mobility aids, with recommendations for future personalization, low light enhancement, and tighter integration with higher level landmark semantics.

Keywords: Landmark Recognition, Computer Vision, TinyML, ESP32-CAM, Assistive Technology, Ultrasonic Sensing

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LIST OF ABBREVIATIONS

ADC	Analog-to-Digital Converter
AI	Artificial Intelligence
CAM	Camera Module (as in ESP32-CAM)
CNN	Convolutional Neural Network
CPU	Central Processing Unit
ETA	Electronic Travel Aid
FPS	Frames Per Second
IoT	Internet of Things
KB	Kilobyte
LDO	Low-Dropout (Voltage Regulator)
LED	Light-Emitting Diode
Li-ion	Lithium-ion (Battery)
LKR	Sri Lankan Rupee (Currency)
MB	Megabyte
ML	Machine Learning
MP	Megapixel
MTBF	Mean Time Between Failures
PCB	Printed Circuit Board
PWM	Pulse-Width Modulation
RAD	Rapid Application Development
RAM	Random Access Memory
SPI	Serial Peripheral Interface (a communication protocol)
TFLM	TensorFlow Lite for Microcontrollers
TinyML	Tiny Machine Learning (ML on microcontrollers)
USB-C	Universal Serial Bus Type-C (connector)

1. INTRODUCTION

The independence, safety, and confidence of visually impaired individuals during mobility are conditioned by timely awareness of immediate physical hazards and understanding of surrounding environmental context. Traditional aids—primarily the white cane—offer reliable tactile feedback but only upon physical contact, leaving a reaction gap for elevated obstacles, recessed drop-offs, narrow or cluttered passages, and overhead barriers. Single-sensor electronic travel aids (ETAs) have attempted to bridge this gap using ultrasound or infrared; however, they frequently produce false alarms or ambiguous signals, eroding user trust and adoption rates [1], [2].

The current group project addresses environmental semantic understanding through a Landmark Recognition Component (e.g., crosswalks, entrances, signals). This proposal focuses on the complementary near-field safety layer: a Multi-Modal Obstacle & Gap Detection Component that fuses dual ultrasonic sensors and embedded vision with lightweight machine learning (TinyML) to classify navigability, detect obstacles, overhead hazards, and drop-offs, and convey graded haptic feedback.

Single-modality distance thresholds misinterpret reflective surfaces, soft materials, and transient occlusions, while pure vision approaches on microcontrollers degrade in low light and impose higher power costs. Fusion of heterogeneous modalities combined with uncertainty-aware decision logic promises improved robustness, reduced false positives, and actionable feedback relevance [3], [4]. Emerging TinyML frameworks now enable inference within strict energy and memory budgets, opening pathways for on-device classification without privacy risks of cloud streaming [5].

A rigorous research approach—covering dataset design, feature engineering, model compression, sensor fusion policy, haptic interaction design, evaluation (technical and user-centered), and commercialization feasibility—will ensure the component advances beyond an engineering prototype into a research-contributive artifact. [15]

1.1 Background & Literature Survey

1.1.1 Traditional mobility aids and limitations

The white cane remains the gold standard due to low cost, simplicity, and high reliability for ground-level detection. Limitations include lack of pre-contact warning, inability to detect silent overhangs (e.g., signboards), difficulty measuring gap width before engagement, and minimal vertical semantic information [2]. Guide dogs enhance navigation but entail training overhead, cost, and limited scalability.

1.1.2 Ultrasonic sensing in assistive devices

Ultrasound offers precise short-range measurements (≈2–400 cm) with millisecond response and low power draw [6]. Limitations: beam divergence reduces lateral resolution; specular reflections from glass/metal cause dropouts; soft textile surfaces absorb sound; multipath echoes produce unstable readings. Multi-transducer arrays improve coverage but raise crosstalk risk if not time multiplexed.

1.1.3 Embedded vision for near-field hazard detection

Low-resolution cameras paired with compressed CNNs can recognize structural blockages, overhead objects, and texture transitions (e.g., approaching staircase). Challenges include lighting variability, motion blur during user gait, dynamic backgrounds (crowds), and the compute-energy trade-off on microcontrollers [7], [8].

1.1.4 Sensor fusion and multi-modal perception

Fusion attempts to integrate complementary strengths: ultrasound's precise scalar range with vision's semantic and spatially distributed cues. Late fusion (decision-level), early fusion (feature concatenation), and hybrid gating strategies reduce failure modes (e.g., false echoes, visual noise). Confidence weighting and Bayesian or heuristic gating can reduce spurious alerts [4], [9].

1.1.5 Edge AI and TinyML enablement

Advances in model quantization, pruning, and operator fusion enable sub-50 KB neural networks executing in <30 ms on ESP32-class hardware, eliminating dependence on network connectivity and mitigating privacy concerns [5]. TensorFlow Lite for Microcontrollers (TFLM) supports static memory arenas and int8 pipelines reducing RAM footprint.

1.1.6 Haptic feedback design in assistive technology

Effective haptic channels require distinctive, low-ambiguity patterns with limited frequency to reduce cognitive and sensory fatigue. Prior studies highlight pattern differentiation (pulse length and rhythm), graduated intensity for urgency, and avoidance of constant vibration to maintain interpretability and trust [10].

1.1.7 Related smart cane systems

Existing solutions: (a) ultrasonic-only threshold devices (binary alerts), (b) vision prototypes offloaded to smartphones (latency, privacy issues), (c) cloud-assisted recognition (connectivity dependency), and (d) multi-sensor experimental platforms lacking quantified ML and uncertainty gating. Few integrate explicit navigable gap classification with overhead and drop-off detection on a single embedded platform at low cost.[12]

1.2 Research gap

Despite progress in electronic travel aids and embedded sensing, several key limitations remain in current obstacle detection and mobility support systems for visually impaired individuals:

1.2.1 Limited robustness of single-modality sensing

Most current smart canes rely solely on ultrasonic sensing or camera-based vision. Ultrasonic-only systems misinterpret reflective surfaces (glass, metal) and soft absorptive materials, while vision-only approaches suffer in low light, motion blur,

and dynamic environments. Relying on a single modality leads to frequent false positives or missed hazards, undermining user trust and adoption.[13]

1.2.2 Absence of navigable gap classification

Existing devices primarily alert users to the presence of obstacles but do not provide meaningful assessment of gap width or safe passage between obstacles. This omission is critical, as visually impaired individuals require clear information about whether narrow or cluttered pathways are traversable.

1.2.3 Overhead and drop-off hazard detection gaps

Many assistive systems emphasize ground-level obstacles but neglect silent overhead hazards (signboards, branches) and recessed drop-offs (stairs, curbs). The lack of multi-plane awareness limits real-world safety, especially in unfamiliar environments.

1.2.4 Dependence on external devices or cloud

Several vision-based prototypes offload processing to smartphones or cloud servers. This creates latency, raises privacy risks, and makes the system unreliable in areas with poor connectivity. True portability and independence require fully embedded, on-device processing.

1.2.5 Underutilization of TinyML for real-time inference

Although advances in TinyML allow highly compressed neural networks to run on microcontrollers with <100 KB memory, few assistive technologies have adopted this approach. Most remain tied to resource-heavy models or external hardware, missing opportunities for low-power, affordable deployment.

1.2.6 Inadequate haptic feedback design

Current systems often rely on binary vibrations (on/off) or constant buzzing, which quickly causes sensory fatigue and fails to convey the type or urgency of hazards. There is limited research on graded, intuitive haptic feedback that differentiates between obstacles, gaps, and urgency levels.

1.2.7 Limited real-world validation and scalability

Most prototypes are tested in controlled lab conditions with small participant groups. Few undergo rigorous user-centered evaluation in diverse environments, and even fewer consider cost, durability, and manufacturing scalability—factors essential for practical deployment.

Below provides a comparison of existing smart cane systems with the proposed research, showing how current solutions address or fail to address these critical gaps.

Table 1.1 Comparison of existing smart stick systems with the proposed system

Feature / System	We WALK Smart Cane (Research 1)	Smart Cane IIT Delhi (Research 2)	IoT-Based Smart Stick (Research 3)	A IoT Smart Stick (Research 4)	Proposed Research
Real-time Multi-Modal Fusion (Ultrasonic + Vision)	✕	✕	✕	✕	✓
Overhead & Drop-off Hazard Detection	✕	✓ (partial ground detection)	✕	✕	✓
Embedded Processing (TinyML)	✕ (smartphone-based)	✓	✕ (cloud/IoT dependent)	✕ (cloud-assisted)	✓
Multimodal Feedback (graded haptic)	✓ (audio + limited haptic)	✓ (vibration only)	✓ (audio only)	✓ (audio only)	✓ (graded haptic)
Portable / Integrated	✓	✓	✓	✓	✓
Affordable / Scalable	✕ (premium cost)	✓ (low-cost design)	✓ (moderate cost)	✕ (IoT infrastructure required)	✓

1.3 Research problem

Visually impaired individuals face persistent challenges in detecting obstacles, overhead hazards, and safe passage gaps during independent mobility. Traditional mobility aids such as white canes and guide dogs provide tactile feedback and ground-level detection but fail to deliver pre-contact warnings, identify navigable gaps, or recognize overhead and recessed hazards. Current electronic travel aids (ETAs), although incorporating ultrasonic or vision sensors, remain limited by false positives, single-modality weaknesses, and inadequate feedback design, thereby restricting user trust and widespread adoption.

The absence of a lightweight, embedded, multi-modal obstacle and gap detection system creates several critical problems:

1.3.1 Safety risks from undetected hazards

Without reliable detection of overhead barriers (e.g., signboards, branches) and recessed drop-offs (e.g., curbs), visually impaired individuals are exposed to injuries that traditional canes cannot prevent.

1.3.2 Inability to evaluate navigable gaps

Existing aids can indicate the presence of obstacles but fail to classify whether narrow passages are traversable. This forces users into trial-and-error navigation, increasing stress and reducing efficiency.

1.3.3 High false alarms and low trust

Ultrasonic-only systems often misinterpret reflective or absorptive surfaces, while vision-only systems degrade in poor lighting or dynamic conditions. Frequent false alarms reduce user confidence and adoption rates.

1.3.4 Dependence on external devices

Several vision-based prototypes rely on smartphones or cloud offloading, introducing latency, privacy concerns, and connectivity limitations, which reduce practicality for everyday use.

1.3.5 Ambiguous feedback mechanisms

Binary or constant vibration alerts used in many ETAs create cognitive overload and do not communicate the type or urgency of hazards. This ambiguity hinders decision-making during real-world mobility.

1.3.6 Lack of user-centered and scalable solutions

Many prototypes remain at the lab stage, tested under limited conditions without considering real-world usability, affordability, and long-term scalability. This limits their impact as viable assistive solutions.

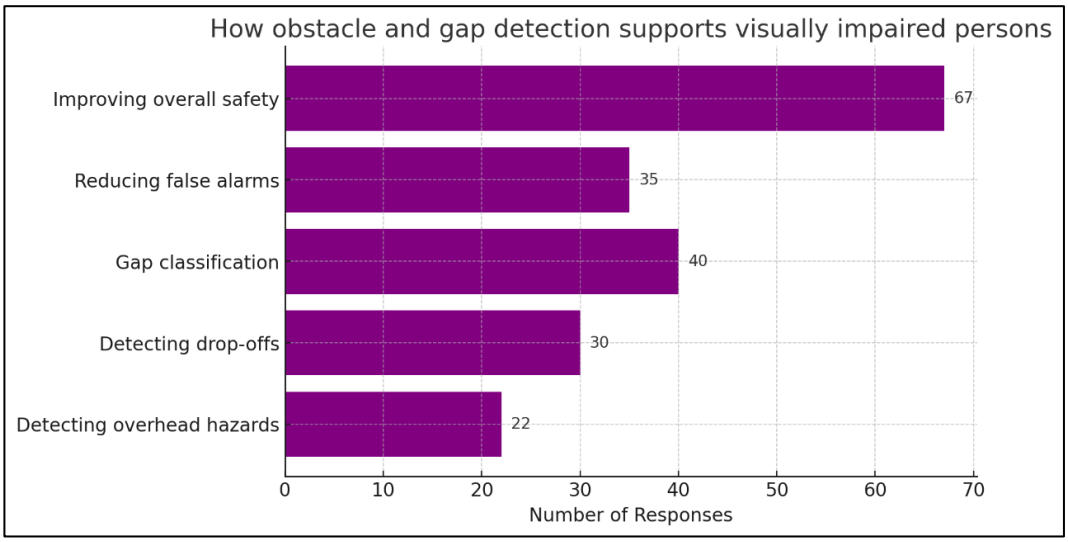

Figure 1.1 Obstacle detection for visual impaired persons classification

2. OBJECTIVES

2.1 Main objective

To develop an intelligent, embedded multi-modal obstacle and gap detection system using ESP32-class microcontrollers, dual ultrasonic sensors, and embedded vision with TinyML, providing real-time detection of obstacles, overhead hazards, drop-offs, navigable gaps, and delivering graded haptic feedback to enhance mobility safety and confidence for visually impaired individuals.

2.2 Specific objectives

2.2.1 Sensor fusion and obstacle detection

The first objective is to integrate dual ultrasonic sensors with an ESP32-CAM vision module to enable multi-plane obstacle detection. To achieve reliable results, sensor fusion algorithms, whether late, early, or hybrid approaches—will be developed to improve detection accuracy and minimize false positives. The system will classify obstacles according to their type, including ground-level, overhead, or recessed hazards, as well as their proximity to the user. Validation of detection performance will be conducted under diverse real-world conditions, both indoors and outdoors, and across varying lighting environments.

2.2.2 Navigable gap assessment

Another objective is to design algorithms capable of determining safe passage widths between detected obstacles. These algorithms will classify gaps into categories such as narrow, medium, or wide, providing visually impaired users with actionable information to support safer navigation. The performance of gap classification will be tested in cluttered and dynamic environments to ensure reliability under realistic use cases.

2.2.3 TinyML model development

A further objective involves the development of lightweight neural networks trained for both obstacle classification and gap assessment, with a target accuracy of at least 85%. To ensure deployment feasibility on the ESP32 platform, model optimization techniques such as quantization and pruning will be applied to reduce the memory footprint. Additionally, the TinyML models will be evaluated for robustness under diverse environmental conditions, including low-light scenarios and dynamic background changes.

2.2.4 Embedded system implementation

The next objective is to implement the complete embedded system on the ESP32 microcontroller, enabling real-time acquisition and processing of sensor data. TinyML inference pipelines will be integrated to perform low-latency obstacle and gap classification, ensuring responsiveness during navigation. The embedded implementation will be optimized to achieve high processing efficiency while minimizing both power consumption and latency, making the system suitable for portable, battery-powered operation.

2.2.5 Haptic feedback system development

The final objective is to design and implement a haptic feedback system that translates detected hazards and navigation cues into graded vibrotactile patterns. Feedback intensity, pulse length, and rhythm will be mapped to hazard type, proximity, and navigable gap width, ensuring clear and intuitive communication with the user. Pilot testing will be conducted with visually impaired individuals to evaluate comprehension, usability, and the overall effectiveness of the feedback system.

2.2.6 System integration and evaluation

This objective focuses on integrating the sensors, TinyML models, and haptic feedback mechanisms into a unified, portable smart cane prototype. The fully integrated system will undergo comprehensive real-world testing to evaluate its accuracy, responsiveness, and robustness in diverse environmental conditions such as varying terrains, lighting levels, and outdoor weather scenarios. In addition, user acceptance testing will be conducted with visually impaired individuals to assess usability, trust, and the extent to which the system enhances overall mobility and independence.

2.2.7 Performance optimization

The final objective is to optimize the system's performance for practical, daily use. Efforts will be directed towards minimizing latency to ensure that feedback is provided in near real-time, thereby supporting timely user decision-making. Energy efficiency will also be improved to enable extended battery life, allowing the device to function reliably throughout the day. Furthermore, the system's durability and reliability will be enhanced to withstand outdoor deployment under different environmental conditions, ensuring consistent performance and user confidence in real-world scenarios.

3. METHODOLOGY

3.1 Overall system description

The overall architecture of the obstacle detection component is designed as a multi-stage pipeline, as illustrated in Figure 3.1. The process begins with simultaneous data acquisition from the hardware sensors. The raw data is then pre-processed, and feature engineering is applied to extract meaningful characteristics. These features are fed into a TensorFlow Lite model running on the ESP32 for classification. The results from the model are processed by a decision and fusion logic module, which also incorporates hysteresis and confidence gating to smooth outputs. Finally, the decided action is translated into a specific haptic feedback pattern delivered to the user.

3.2 Development methodology

The project follows an iterative development approach based on the Rapid Application Development (RAD) methodology, allowing for quick prototyping and continuous refinement based on user feedback and testing results. This RAD-based iterative approach mitigates risk by identifying flaws in design or implementation early, ensures the final product is deeply user-centric, and provides the flexibility to incorporate insights and technological improvements as they arise throughout the project's lifecycle.

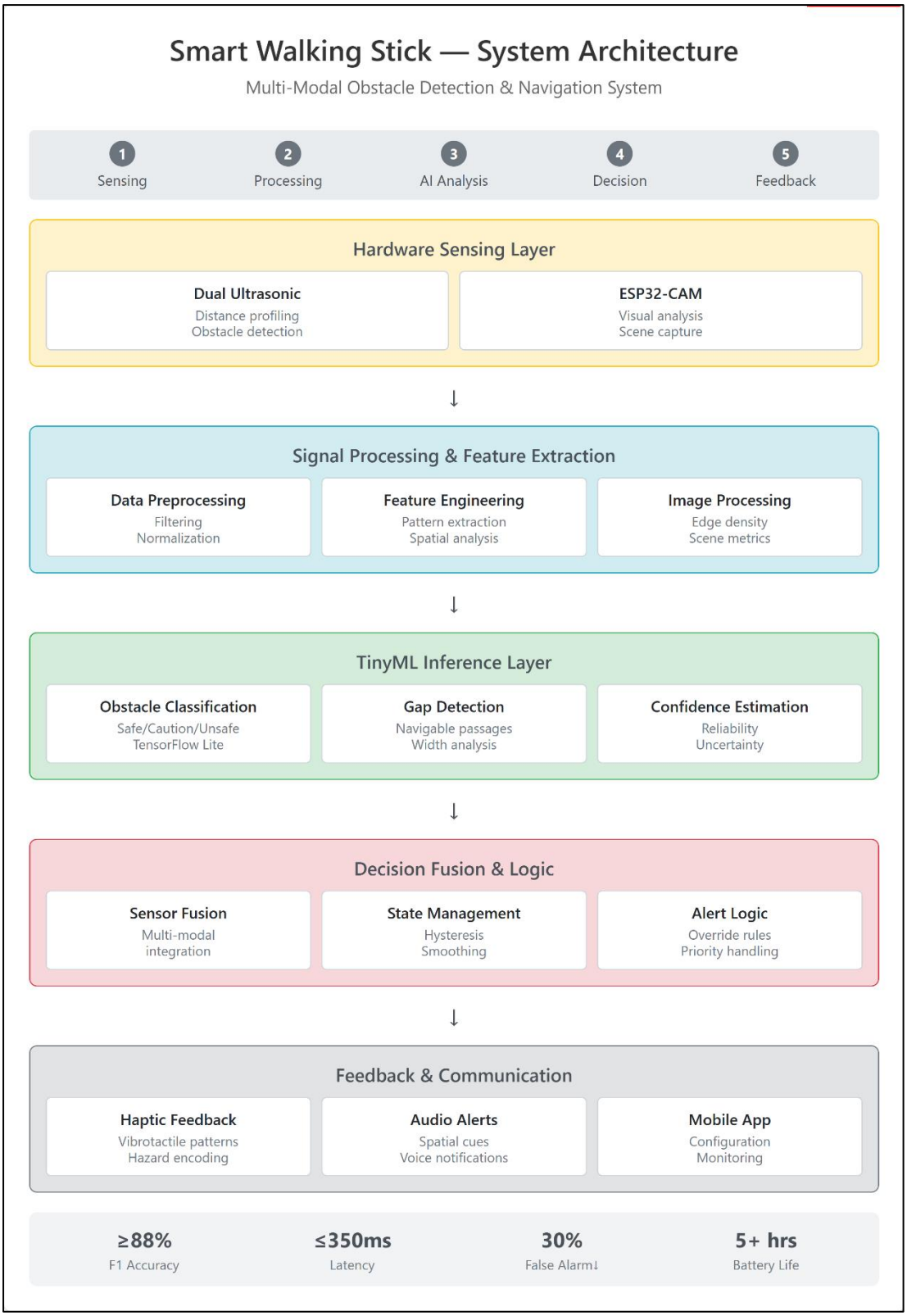

Figure 3.1 Component system architecture

3.2.1 Ghantt Chart

The Gantt chart for the "Smart Walking Stick for Visually Impaired" project outlines a structured timeline from August 2025 to June 2026. It details key phases, including planning, design, and model development, with specific tasks such as project initiation, literature review, hardware/software architecture design, and mobile app development. Each task is color-coded and scheduled to ensure progress, quick prototyping, and continuous refinement based on user feedback and testing results.

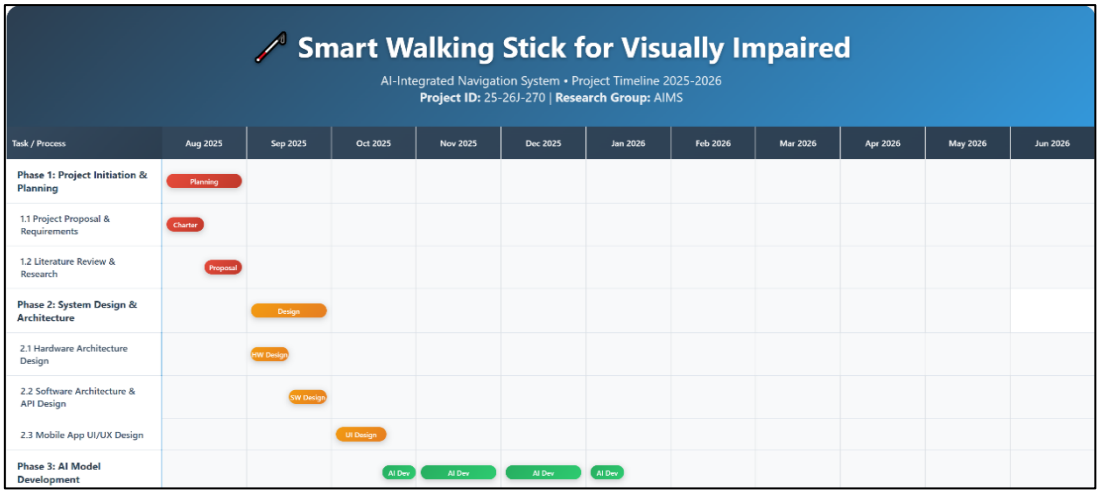

Figure 3.2 Gantt Chart for the entire research project

Full hosted Ghantt chart for the whole year - https://ghant-chart-25-26j-270.netlify.app/

3.2.2 Work Breakdown Structure

Phase 1: Research and Planning (Weeks 1-4)

- Literature review and gap analysis
- Requirements gathering and analysis
- Technical feasibility study
- Hardware component selection
- Project planning and timeline development

Phase 2: System Design (Weeks 5-8)

- System architecture design
- Hardware schematic design
- Software architecture planning
- User interface mockups
- Testing strategy development

Phase 3: Hardware Implementation (Weeks 9-16)

- ESP32-CAM module setup and configuration
- Ultrasonic sensor integration
- Vibrotactile feedback system implementation
- Power management circuit design
- Component assembly and testing

Phase 4: Software Development (Weeks 17-28)

- Embedded software development
- TinyML model development and training
- Sensor fusion algorithm implementation
- Decision logic programming
- Haptic feedback control system

Phase 5: Integration and Testing (Weeks 29-36)

- Hardware-software integration
- Unit testing of individual components
- Integration testing
- System performance testing
- User acceptance testing preparation
- Bug fixing and optimization

Phase 6: Evaluation and Documentation (Weeks 37-40)

- Real-world testing scenarios
- Performance evaluation and metrics collection
- Comparative analysis with existing solutions
- Final documentation
- Presentation preparation
- Project deliverable finalization

3.3 Technical Implementation Approach

3.3.1 Machine Learning Model Development

Dataset Creation and Preparation: The development of robust TinyML models requires comprehensive datasets representing various environmental conditions and obstacle scenarios. Data collection will be conducted in controlled and real-world environments to ensure model generalization.

Data Collection Strategy:

The data collection process will be carried out across a wide range of environments to ensure system robustness. Indoor datasets will be gathered from locations such as offices, homes, shopping malls, and corridors, while outdoor datasets will include streets, sidewalks, parks, and parking areas. To test adaptability under varying illumination, data will be collected in multiple lighting conditions including bright sunlight, overcast skies, twilight, and artificial lighting. Furthermore, the system will be evaluated under diverse weather conditions such as clear, rainy, and foggy scenarios. A variety of obstacle types will be incorporated, covering static obstacles like poles, walls, and furniture; dynamic obstacles such as people and vehicles; overhead hazards including branches and signboards; and ground-level hazards such as stairs and curbs.

Feature Engineering:

Feature engineering will focus on extracting meaningful information from both ultrasonic and visual data streams. From ultrasonic sensors, features such as distance symmetry, variance, slope trends, and stability metrics will be computed to capture object positioning and motion consistency. Visual features will be derived from camera inputs, including edge density, scene occupancy, brightness levels, and texture analysis, enabling richer environmental understanding. Fusion features will then be designed to combine both modalities, with emphasis on confidence correlation, temporal consistency, and spatial alignment, thereby strengthening the reliability of obstacle detection and classification

Model Architecture:

A lightweight Convolutional Neural Network (CNN) will be developed to meet the computational and memory constraints of the ESP32 platform. The input layer will process multi-modal sensor data, combining both ultrasonic and visual features. Convolutional layers using depthwise separable convolutions will extract salient features efficiently, while pooling layers will reduce spatial dimensions and computational complexity. Dense layers will then perform classification and confidence estimation, with the output layer providing a three-level safety classification: *Safe*, *Caution*, or *Unsafe*, each accompanied by confidence scores.

Model Optimization:

To ensure deployment feasibility on the ESP32 microcontroller, multiple optimization techniques will be applied. Quantization, specifically INT8 quantization, will be used to reduce model size and improve inference speed. Pruning methods will eliminate redundant neural connections, minimizing memory footprint without sacrificing accuracy. Additionally, knowledge distillation will be employed to transfer essential learning from larger, more complex teacher models to the lightweight embedded model, ensuring strong performance under real-world constraints.

3.3.2 ESP32-CAM Implementation

Sensor Integration:

The hardware integration of the smart cane system will combine multiple sensing and feedback components into a cohesive platform. A forward-facing ultrasonic sensor (HC-SR04 or equivalent) will serve as the primary obstacle detection unit, while a secondary ultrasonic sensor, angled upwards, will detect overhead hazards. Vibrotactile motors will be integrated at multiple positions to provide directional haptic feedback, enabling the user to distinguish left, right, and central hazards. Power management will be handled through an embedded battery monitoring and charging circuit, ensuring reliable operation and sustained portability.

Real-time Processing Pipeline:

The system will follow a structured real-time data processing pipeline. First, sensor data acquisition will involve synchronized readings from both the ultrasonic sensors and the onboard camera. This raw data will undergo preprocessing, including noise filtering, calibration, and normalization to ensure consistent quality. Feature extraction will then be performed in real time, computing both ultrasonic and visual features as defined in the feature engineering stage. Model inference will be carried out using the TinyML neural network to classify safety states. A decision fusion layer will combine the results of multiple modalities, applying confidence weighting to strengthen reliability. Finally, output generation will trigger the appropriate haptic feedback pattern, providing timely and intuitive navigation cues to the user.

3.3.3 Communication System

The communication architecture will employ multiple protocols to ensure reliable interaction between components. The I2C protocol will manage data transfer between the ESP32 microcontroller and peripheral sensors, while the SPI protocol will support high-speed transfer of camera data. General-purpose input/output (GPIO) pins will control the vibrotactile motors, and analog-to-digital converter (ADC) channels will monitor battery voltage for power management. For system diagnostics and monitoring, optional data logging will be supported through local SD card storage. In addition, Wi-Fi connectivity will enable wireless monitoring and debugging, offering flexibility during testing and future extensions of the system.

3.3.4 Feedback System Design

Obstacle Detection Patterns:

The haptic feedback system will employ distinct vibration patterns to communicate obstacle types to the user. Ground-level obstacles will be signaled by a single short pulse lasting 200 milliseconds, while overhead hazards will be indicated through a double-pulse pattern consisting of a 200-millisecond vibration, a 100-millisecond pause, followed by another 200-millisecond vibration. Drop-off hazards, such as stairs or curbs, will be conveyed using continuous vibrations with gradually increasing

intensity to emphasize urgency. In contrast, a clear path will either result in no vibration feedback or, optionally, a very brief confirmation pulse of 50 milliseconds.

Urgency Levels:

The system will also encode the urgency of obstacles based on proximity, allowing users to gauge the level of caution required. At safe distances, no feedback or only minimal confirmation pulses will be delivered, ensuring that the user is not overwhelmed by unnecessary signals. When an obstacle enters the caution range, gentle pulses will begin, gradually increasing in frequency as the user moves closer. In situations of immediate danger, the system will trigger rapid and strong vibrations at maximum intensity, alerting the user to take urgent action.

Gap Width Communication:

In addition to obstacle alerts, the system will communicate gap width information to assist with safe passage decision-making. A narrow but passable gap will be indicated by a single long pulse of 500 milliseconds, while medium gaps will be represented by two medium-length pulses of 300 milliseconds each. Wide gaps, signaling a clear and easy passage, will be conveyed using three short pulses of 150 milliseconds each.

Motor Configuration:

To enhance clarity and directional awareness, the system will integrate multiple vibrotactile actuators positioned on the left, right, and center of the cane handle or grip. This arrangement will allow directional feedback, enabling the user to identify the location of hazards. Pulse-width modulation (PWM) will be applied to adjust vibration strength dynamically, supporting variable intensity feedback. Additionally, duty cycling techniques will be used to conserve battery life, ensuring that the system remains power-efficient while delivering consistent and effective haptic communication.

3.4 Testing and Validation Methodology

The testing and validation process will follow a structured, multi-phase approach to ensure that the smart cane system is accurate, reliable, energy-efficient, and user-friendly. Performance will be evaluated using a set of quantitative and qualitative metrics. Accuracy will be assessed through precision, recall, and F1-score for obstacle classification. Latency will be measured as the end-to-end response time between sensor input and haptic feedback output, ensuring near real-time performance. Reliability will be evaluated by analyzing false positive and false negative rates, while energy efficiency will be monitored through power consumption and battery life profiling. Usability will be measured by assessing user comprehension and interpretation of the vibrotactile feedback patterns.

Testing Phases:

Phase 1: Component Testing

The first phase will focus on validating individual system components. Ultrasonic sensors and the ESP32-CAM will be tested for accuracy and reliability under controlled conditions. The TinyML model will be validated using benchmark datasets to confirm obstacle classification accuracy. User trials will also be conducted to evaluate recognition of haptic feedback patterns. Additionally, power consumption will be profiled to establish baseline energy requirements.

Phase 2: Integration Testing

The second phase will assess system-level performance after integrating all components. Sensor fusion algorithms will be validated to ensure correct multi-modal operation. Real-time processing performance and system latency will be measured to confirm low-latency response. Multi-modal failure modes will also be analyzed to ensure that the system can maintain safe functionality even when one or more sensing modalities are degraded.

Phase 3: Environmental Testing

The third phase will evaluate system performance in real-world environments. Indoor testing will be carried out in settings such as corridors, shopping malls, and homes, while outdoor testing will be performed on sidewalks, parks, and streets. The system will also be tested under various lighting conditions, including bright sunlight, dim environments, and artificial lighting, to ensure robustness under diverse operational scenarios.

Phase 4: User Acceptance Testing

The final phase will involve visually impaired participants, who will test the system in realistic navigation tasks. This phase will evaluate usability, effectiveness, and overall mobility enhancement compared to existing mobility aids. Feedback will be collected from participants to refine the system design and improve user trust and satisfaction.

Validation Criteria:

The system will be considered validated if it achieves a minimum safety classification accuracy of 88% F1-score. Additionally, false alarms must be reduced by at least 30% compared to an ultrasonic-only baseline. The end-to-end response latency must not exceed 350 milliseconds at the 95th percentile, ensuring timely feedback. Finally, the device must demonstrate a minimum battery life of five hours during continuous operation to meet daily usability requirements.

4. PROJECT REQUIREMENTS

4.1 Functional requirements

Table 4.1 Functional requirements table

ID	Requirement	Priority	Description
FR1	Real-time obstacle detection	High	System shall detect obstacles within 0.5-3m range using ultrasonic sensors
FR2	Visual hazard recognition	High	ESP32-CAM shall identify overhead and ground-level hazards using computer vision
FR3	Gap width assessment	High	System shall classify navigable gaps as narrow, medium, or wide
FR4	Multi-modal sensor fusion	High	Combine ultrasonic and visual data for improved accuracy
FR5	TinyML inference	High	Execute lightweight neural network models on ESP32 hardware
FR6	Haptic feedback generation	High	Provide distinct vibrotactile patterns for different hazard types
FR7	Power management	Medium	Monitor battery status and implement power-saving modes
FR8	Data logging	Low	Optional logging for debugging and performance analysis

4.1.1 Core Functionality

Obstacle Detection and Classification:

The system is designed to provide reliable, real-time detection and classification of obstacles for visually impaired users. It can identify static obstacles such as walls, poles, and furniture within a range of 0.5 to 3 meters. Overhead hazards, including low-hanging branches, signboards, and barriers, are also detected to prevent potential collisions. Ground-level hazards such as stairs, curbs, potholes, and elevated surfaces are recognized to reduce the risk of trips or falls. The system can distinguish between solid obstacles and passable gaps, allowing users to make informed navigation

decisions. Each detected obstacle is classified according to urgency levels: immediate, caution, or safe, ensuring that users receive timely guidance.

Gap Assessment:

In addition to obstacle detection, the system performs precise gap assessment between obstacles. Gap widths are measured with an accuracy of ±10 cm and classified as narrow (<60 cm), medium (60–90 cm), or wide (>90 cm). These classifications consider the user's shoulder width and mobility requirements, ensuring that the navigation advice is personalized and practical. Confidence levels are also provided for each gap assessment to support informed decision-making while moving through narrow or crowded spaces.

Environmental Adaptation:

The system dynamically adapts to different environmental conditions to maintain reliable performance. Sensitivity adjustments are made based on ambient lighting conditions, while compensation mechanisms address sensor performance variations caused by weather effects such as rain or fog. The system also adapts to different walking speeds and individual gait patterns, providing consistent feedback regardless of user motion. Furthermore, it is capable of handling dynamic environments with moving obstacles, ensuring robust real-world operation across diverse indoor and outdoor scenarios.

4.1.2 User Interface Requirements

Haptic Feedback Interface:

The system's user interface relies on a haptic feedback mechanism to provide intuitive navigation cues for visually impaired users. It generates distinct vibrotactile patterns for each type of hazard, allowing users to differentiate between ground-level obstacles, overhead hazards, and drop-offs. Multiple intensity levels, ranging from three to five, are supported to convey varying degrees of urgency and proximity. Directional feedback is provided through actuators positioned to indicate left, right, center, or all directions, helping users accurately localize obstacles in their path. Additionally, confirmation vibration patterns are included to indicate successful navigation or safe

passage, reinforcing user confidence and ensuring a clear understanding of the system's guidance.

4.1.3 Communication Requirements

The system requires reliable communication both within its components and with external devices. For inter-component communication, the ESP32 microcontroller interfaces with sensor modules using the I²C protocol, while the SPI interface handles high-speed data transfer from the camera. GPIO pins are used to control the vibrotactile actuators, and ADC channels monitor battery levels and sensor status to ensure safe and efficient operation.

For external communication, the system supports Wi-Fi connectivity, enabling debugging, system monitoring, and data collection during testing or deployment. Additionally, Bluetooth Low Energy (BLE) is implemented to allow integration with smartphones, providing users with the ability to configure system settings, access logs, or receive supplementary navigation assistance. These communication capabilities ensure seamless interaction both within the device and with external support systems.

4.2 Non-Functional Requirements

Table 4.2 non-functional requirements

ID	Requirement	Priority	Specification
NFR1	Response Time	Critical	≤350ms end-to-end latency at 95th percentile
NFR2	Accuracy	Critical	≥88% F1-score for safety classification
NFR3	Battery Life	High	≥5 hours continuous operation
NFR4	False Alarm Rate	High	≤30% compared to ultrasonic-only baseline
NFR5	Operating Temperature	High	-10°C to +50°C operational range

NFR6	Weight	High	≤200g additional weight to standard cane
NFR7	Durability	Medium	IP54 rating for dust and splash resistance
NFR8	Memory Usage	High	≤512KB RAM, ≤4MB Flash storage
NFR9	Processing Efficiency	High	≤80% CPU utilization during normal operation
NFR10	Cost Effectiveness	Medium	Total hardware cost ≤$150 USD

4.2.1 Performance Requirements

Real-time Processing:

The system is required to operate with real-time responsiveness to ensure effective and safe navigation for users. The complete cycle from sensor data acquisition to haptic feedback must be completed within 350 milliseconds, providing near-instantaneous guidance. Sensor data should be processed at a minimum frequency of 10 Hz to maintain smooth and continuous environmental monitoring. To manage thermal performance, the average CPU utilization must remain below 80%, preventing overheating during extended operation. Memory usage must be optimized to operate within the 512 KB SRAM constraints of the ESP32, ensuring efficient processing without exceeding hardware limitations.

Accuracy and Reliability:

The system is designed to deliver high levels of accuracy and reliability in obstacle detection for visually impaired users. It must achieve a minimum F1-score of 88% when classifying obstacles versus safe zones, ensuring dependable navigation guidance. False positive rates should be limited to 15% or lower in controlled environments, reducing unnecessary alerts, while the false negative rate for safety-critical obstacles must not exceed 5% to maintain user safety. Additionally, confidence calibration is required so that uncertainty estimates remain within ±10% of empirical accuracy, providing users with trustworthy and precise feedback during navigation.

Sensor Performance:

The system's sensors are designed to provide reliable and robust environmental perception for safe navigation. Ultrasonic sensors must detect obstacles accurately within a range of 0.5 to 3.0 meters, while the vision module should provide effective visual processing for approximately 1.5 meters forward. The ultrasonic sensors are required to cover an angular range of ±30° to ensure adequate spatial awareness. Additionally, the system must maintain consistent performance across various lighting conditions, including bright sunlight, artificial indoor lighting, and low-light environments, to guarantee dependable operation in diverse real-world scenarios.

4.2.2 Reliability Requirements

System Availability:

The system must ensure high operational reliability to maintain user safety and trust. It is required to achieve at least 99.5% uptime during active use, providing consistent availability for navigation assistance. The mean time between failures (MTBF) should be at least 1,000 hours of operation, reflecting long-term dependability. Additionally, the system must include fault tolerance mechanisms, allowing for graceful degradation if individual sensors fail, so that partial functionality is maintained and users remain informed of potential hazards.

Data Integrity:

To ensure reliable operation, the system incorporates robust data integrity mechanisms. Automatic sensor calibration and drift compensation maintain accuracy over time, while built-in diagnostics detect sensor malfunctions promptly. In the event of system errors, a fail-safe mode is activated, defaulting to maximum caution to ensure user safety.

Environmental Resilience:

The system is also designed to operate reliably under diverse environmental conditions. It functions effectively within a temperature range of 10°C to 40°C and tolerates relative humidity levels up to 85%. The device can withstand typical walking and handling vibrations, and its housing provides dust and splash protection with an IP54 rating, enabling safe outdoor use even in moderately challenging environments.

4.2.3 Portability and Integration Requirements

Physical Constraints:

The smart cane system is designed with portability and seamless integration in mind. Its physical design adds no more than 200 grams to a standard white cane and maintains a compact form factor that does not interfere with normal cane use. The attachment system securely mounts the device without requiring permanent modifications, while preserving the balance and handling characteristics of the cane to ensure natural and comfortable operation.

Power Requirements:

Power requirements include a rechargeable Li-ion battery capable of supporting at least five hours of active use, with an integrated protection circuit for safety. LED indicators provide real-time information on battery status and overall system operation.

Environmental Adaptability:

The system is also environmentally adaptable, performing effectively in indoor environments such as offices, homes, and retail spaces, as well as in outdoor urban and suburban settings. It automatically adjusts to varying lighting conditions, from bright sunlight to low-light situations, and maintains reliable operation in light rain and windy conditions, ensuring consistent functionality across diverse real-world scenarios.

4.3 System Requirements

4.3.1 Hardware Requirements

Table 4.3 Hardware Requirements Table

Component	Specification	Quantity	Purpose
ESP32-CAM	Dual-core 240MHz, 520KB SRAM	1	Main processing unit
OV2640 Camera	2MP, UXGA resolution	1	Visual hazard detection
Ultrasonic Sensor	HC-SR04 or equivalent	2	Distance measurement
Vibrotactile Motors	3V, adjustable intensity	3	Haptic feedback
Li-ion Battery	3.7V, 3000mAh minimum	1	Power supply
Charging Circuit	USB-C, 5V input	1	Battery management
Voltage Regulator	3.3V output, low dropout	1	Power regulation

Hardware Specifications

The smart cane system relies on a carefully selected hardware setup to ensure accurate sensing, reliable performance, and efficient power management. Power regulation is provided by a low-dropout voltage regulator delivering a stable 3.3V output, supporting the entire electronics system.

Ultrasonic sensors form a primary component of the obstacle detection system, offering a range from 2 cm to 300 cm with ±3 mm accuracy. These sensors operate at

40 kHz and feature a narrow beam angle of less than 15°, providing precise spatial detection of obstacles. The camera module complements ultrasonic sensing, providing up to 2MP resolution (1600×1200) and an adjustable frame rate of 10–30 FPS. The camera's field of view spans 66° diagonally, and it is optimized for enhanced low-light performance, ensuring effective indoor and outdoor operation.

The power system consists of a rechargeable battery with a minimum capacity of 3,000 mAh, supporting more than five hours of continuous operation. Full charging is achieved in under four hours, with integrated power management circuitry providing charging protection and battery monitoring. Low-power modes are implemented to extend standby operation and maximize overall energy efficiency.

4.3.2 Software Requirements

Table 4.4 Software Requirements Table

Software Component	Version/Specification	Purpose
Arduino IDE	v1.8.x or newer	Development environment
ESP32 Arduino Core	v2.0.x or newer	Hardware abstraction
TensorFlow Lite	For Microcontrollers	ML model inference
OpenCV (micro)	Lightweight version	Image processing

Development Environment:

The development of the smart cane system utilizes the Arduino IDE as the primary integrated development environment, with ESP32 board support for programming and deployment. For advanced development workflows, PlatformIO serves as an alternative IDE, providing enhanced project management and debugging capabilities. Source code management is handled using Git, ensuring version control and collaborative development. Documentation is generated with Doxygen to maintain clear, structured code references. The software architecture also incorporates timers

and exception handling to manage real-time processing and system reliability effectively.

Machine Learning Framework:

For the machine learning component, TensorFlow Lite for Microcontrollers is employed as the model inference engine, enabling lightweight and efficient execution on the ESP32. The models are quantized in INT8 format to minimize memory usage, with a footprint of less than 100 KB for storage. Inference is optimized to complete in under 50 milliseconds per classification, supporting near real-time obstacle detection and navigation feedback.

5. BUDGET AND BUDGET JUSTIFICATION

5. 1 Hardware Costs

Item	Description	Unit Cost (LKR)	Quantity	Total Cost (LKR)
ESP32-CAM Module	Main microcontroller with camera	2550	1	2550
Ultrasonic Sensor	HC-SR04 distance sensor	240	2	480
Vibrotactile Motors	3V coin vibration motors	200	5	1000
Li-ion Battery	3.7V 3000mAh rechargeable	1800	1	1800
Enclosure/Housing	Custom 3D printed housing	400	2	800
Mounting Hardware	Screws, brackets, connectors	300	1	300
Wires and Connectors	Various electronic components	500	1	500
Tools and Equipment	Soldering, testing equipment	1000	1	1000
Subtotal Hardware				**8430**

5.2 Software and Development Costs

Category	Description	Cost (LKR)	Justification
Arduino IDE	Free development environment	free	Open-source development platform
TensorFlow Lite	Free ML framework	free	Google's open-source ML library
Google Colab	ML model training platform	free	
GitHub	Version control and collaboration	free	Free for academic use
Documentation	Printing, binding, presentation materials	1000	Final documentation and presentations
Subtotal		**1000**	

5.3 Operational and Miscellaneous Costs

Category	Description	Cost (LKR)	Justification
Transportation	Travel for user testing and data collection	1500	Access to diverse testing environments
Internet and Communication	Data plans for testing and research	500	Internet access for development
Unexpected Expenses	10% buffer for unforeseen costs	1000	Risk mitigation for component failures
Subtotal		**3000**	

5.5 Budget Justification

Hardware Investment Rationale: The hardware budget represents the core investment in developing a functional prototype. The ESP32-CAM platform provides excellent value with integrated WiFi, Bluetooth, and camera capabilities at a low cost. Multiple units are budgeted for development, testing, and backup purposes. The ultrasonic sensors and vibrotactile motors are cost-effective components that provide reliable performance for the target application.

Software and Development Justification: The software costs are minimal due to the selection of open-source tools and frameworks. The primary expense is Google Collab Pro for intensive machine learning model training, which provides GPU acceleration at a fraction of the cost of dedicated hardware. This subscription-based model allows for efficient resource utilization during the development phase.

Return on Investment: The project's potential impact on visually impaired individuals' mobility and independence justifies the investment. The research outcomes will contribute to academic knowledge while developing practical solutions for an underserved community. The open-source approach ensures broader accessibility and continued development by the research community.

6. COMMERCIALIZATION

Target Market

The primary target market consists of visually impaired individuals in Sri Lanka and the wider South Asian region, aged between 15 and 80 years, particularly those from middle-income backgrounds. The secondary market includes healthcare institutions, non-governmental organizations, government agencies, and educational centers that support or provide services to the visually impaired.

Value Proposition

The solution offers enhanced safety and independence by proactively detecting multiple types of hazards, including overhead obstacles, thereby reducing accident risks and enabling more confident navigation. Its user-centric design emphasizes intuitive haptic feedback, customizable settings, and privacy-focused offline operation. From a technical perspective, the product integrates cost-effective and energy-efficient TinyML technology that operates without internet connectivity, ensuring reliability in diverse environments.

Competitive Advantage

The competitive edge lies in its unique multi-modal sensor fusion and specialized edge AI designed for gap and overhead detection. Compared to existing alternatives, it provides a significant cost advantage, being 60–70% cheaper, while offering superior performance and higher user acceptance. In addition, the product incorporates proprietary algorithms, haptic patterns, and optimization techniques tailored for embedded systems, strengthening its intellectual property position.

Business Model

The revenue model includes direct sales of devices to individual consumers and institutions, complemented by service contracts and technology licensing opportunities. The cost structure is primarily driven by manufacturing, which accounts for around 40–45%, followed by research and development at approximately 20–25%.

Future Expansion

In the short term (1–2 years), the focus will be on refining the product, optimizing manufacturing processes, and entering new regional markets. In the medium term (3–5 years), the strategy includes expanding the product line, integrating additional connectivity and AI features, and scaling to global markets. In the long term (5+ years), the vision is to develop a broader ecosystem of assistive technologies and establish the company as a global research leader in the field.

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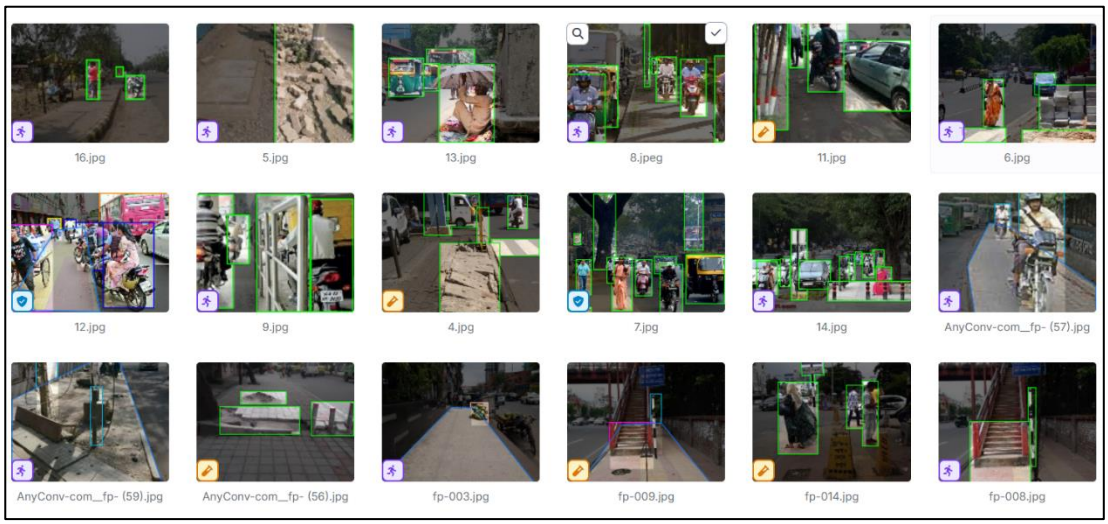
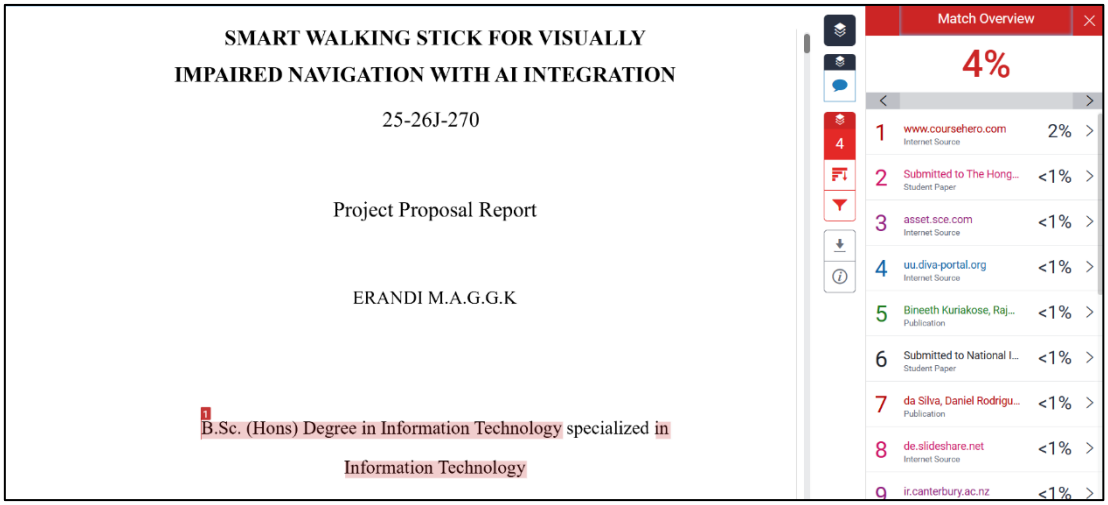
APPENDICES

Sample dataset images:



Plagiarism report (Turnitin)